



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to NAMAKKAL TK LORRY OWNERS ASSN

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

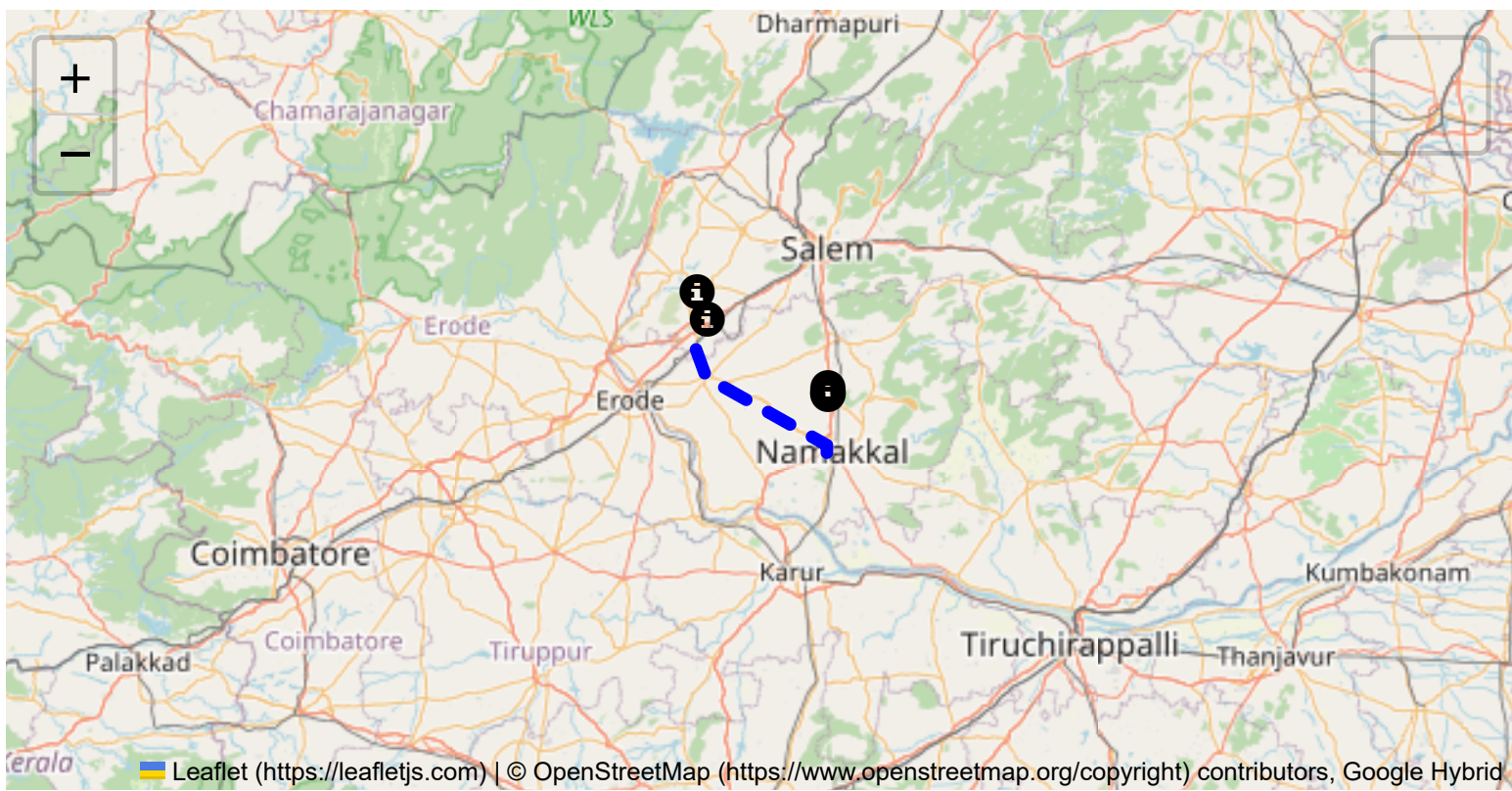
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

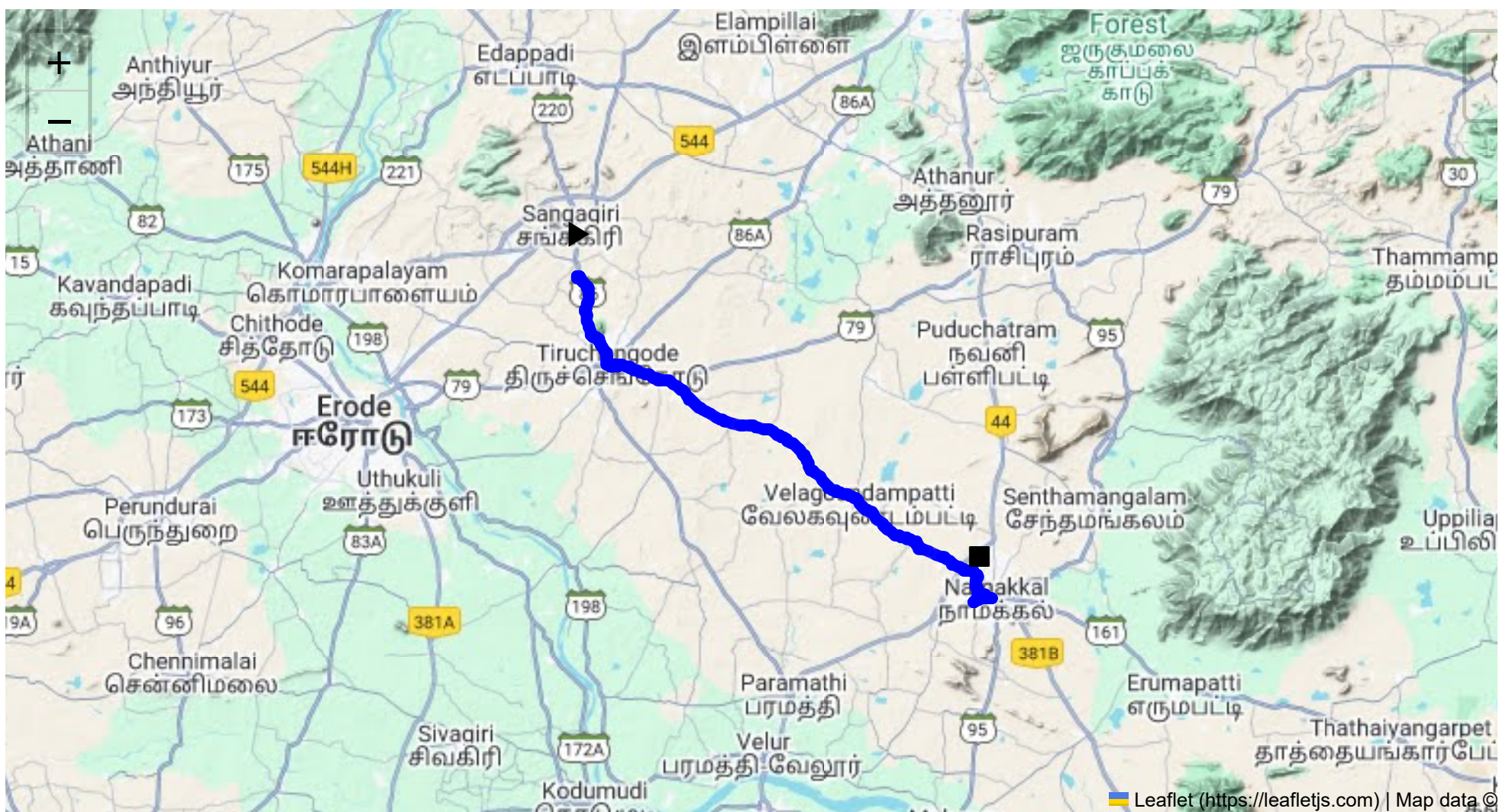
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 46.59 km
Estimated Duration: 1.1 hours
Adjusted Duration (Heavy Vehicle): 1.4 hours
Start: (11.4381, 77.8734)
End: (11.217866, 78.154414)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map: The route from Sangagiri to Namakkal through various points involves primarily regional and state roads. The journey passes through towns like Tiruchengode and Nallipalayam, before entering Namakkal. This stretch is mostly on highways and well-traveled secondary roads, joining major towns and industrial zones.

2. Typical Weather Conditions and Potential Weather-Related Hazards: The region experiences a tropical climate with a hot summer, a significant monsoon season, and a mild winter. During the monsoon (June to September), heavy rains can lead to flash floods and water-logging, impacting road conditions. High temperatures in summer may affect vehicle performance.

3. Analysis of Traffic Patterns: Peak traffic hours are generally between 8:00-10:00 AM and 5:00-7:00 PM. Congestion is likely around key urban areas such as Tiruchengode and Namakkal, especially near market zones and junctions. It is advisable to avoid these areas during peak hours to minimize delays.

4. Assessment of Road Quality and Infrastructure: Road conditions vary. Major highways are generally in good condition, although some rural and connecting roads might have potholes and narrower lanes. Bridges and intersections may require cautious navigation, especially during adverse weather.

5. Suggestions for Alternative Routes for Emergencies: If the primary route is compromised, an alternative is to take State Highway 94 towards Namakkal through Rasipuram, which is usually less congested and provides direct access. Always check real-time traffic updates to assess the best options.

6. Summary of Local Regulations Affecting Hazardous Material Transport: Transport of hazardous materials is regulated with specific guidelines on containment, signage, and movement times, often restricted during peak hours in urban areas. Permits and clearances from local authorities are required prior to departure.

7. Overview of Historical Incidents: While not frequent, incidents involving heavy vehicles occur due to weather, overloaded trucks, and driver fatigue. Collision and overturning have been recorded near Tiruchengode due to sharp turns and inadequate road banking.

8. Environmental Considerations and Sensitive Areas: The route crosses a few agricultural zones and smaller water bodies. Care should be taken to prevent leaks or spills from impacting these sensitive environments. Follow all environmental protection protocols rigorously.

9. Analysis of Communication Coverage: The majority of the route has adequate mobile coverage. Dead zones may be encountered in low-lying or hilly areas, particularly between Tiruchengode and Nallipalayam. Plan communication strategies accordingly.

10. Estimated Emergency Response Times: Emergency response in urban and semi-urban areas like Namakkal is faster (approx. 15-30 minutes), whereas rural segments may experience delayed response times due to distance and road conditions (up to 45 minutes or more).

12. Overall Summary of Risk Assessment: The route presents moderate risk primarily due to weather conditions, traffic congestion in urban areas, and varying road quality. Adequate preparation, compliance with local regulations, and real-time monitoring of conditions can mitigate these risks significantly. Transport companies should ensure drivers are well-trained in emergency procedures and aware of alternative routes. Constant communication and adherence to safety protocols for hazardous material are crucial.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	U-Turn	High	11.2184729, 78.1635718	10 KM/Hr
2	U-Turn	High	11.2164807, 78.1511632	10 KM/Hr
3	Turn	High	11.43811, 77.87348	15 KM/Hr
4	Turn	Medium	11.43956, 77.87340	30 KM/Hr
5	Turn	High	11.43968, 77.87345	15 KM/Hr
6	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
7	Turn	High	11.37868, 77.89498	15 KM/Hr
8	Blind Spot	Blind Spot	11.23446, 78.15458	10 KM/Hr
9	Turn	Medium	11.22243, 78.15379	30 KM/Hr
10	Turn	High	11.22081, 78.15944	15 KM/Hr
11	Blind Spot	Blind Spot	11.21910, 78.15911	10 KM/Hr
12	Turn	High	11.21847, 78.16357	15 KM/Hr
13	Blind Spot	Blind Spot	11.21841, 78.16359	10 KM/Hr
14	Blind Spot	Blind Spot	11.21648, 78.15116	10 KM/Hr
15	Turn	High	11.21655, 78.15114	15 KM/Hr
16	Turn	High	11.21765, 78.15430	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
7	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
11	hospital	Vivekananda Medical Care Hospital	11.364562, 77.9482668	30 km/h	Medium
13	hospital	Manickampalayam, Government Hospital	11.32887, 78.017713	30 km/h	Medium
19	clinic	Rithiksha Physio Care	11.2189918, 78.159957	30 km/h	Medium
20	clinic	Sri Sara ENT Clinic	11.218949, 78.159922	30 km/h	Medium
21	hospital	Namakkal Neurocare Hospital	11.218663, 78.15976	30 km/h	Medium
22	hospital	Vasantham Hospital	11.2187521, 78.161599	30 km/h	Medium
23	hospital	Maharishi Eye Hospital	11.219304, 78.1610484	30 km/h	Medium
24	hospital	Shanmuga Heart Centre	11.22124, 78.163711	30 km/h	Medium
25	clinic	Krishnan Urology Clinic	11.2184523, 78.1643654	30 km/h	Medium
26	hospital	Natasan Hospital	11.2202016, 78.1639129	30 km/h	Medium
27	hospital	Children's Hospital	11.220084, 78.1639673	30 km/h	Medium
29	hospital	Surya Hospital, Namakkal	11.2178506, 78.1660375	30 km/h	Medium
30	hospital	Vairam Hospital, Namakkal	11.217603, 78.153269	30 km/h	Medium
31	clinic	Pranauv Clinic	11.215361, 78.1484232	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
9	college	Vivekanandha Educational Institutions for Women	11.3596633, 77.9435556	30 km/h	Medium
10	college	Vivekanandha Dental College	11.3643705, 77.9475897	30 km/h	Medium
12	college	Vivekanandha College of engineering for Women	11.3565938, 77.9465342	30 km/h	Medium
14	college	The Surabi College of Polytechnic	11.2660726, 78.0939552	30 km/h	Medium
15	college	CMS College Of Engineering	11.2647508, 78.0969044	30 km/h	Medium
16	school	Porasapalayam Government School	11.2504918, 78.1163024	30 km/h	Medium
17	school	Sri Vinayaga Higher Secondary School	11.2479837, 78.1182908	30 km/h	Medium
18	school	Children's Park School	11.2237731, 78.159543	30 km/h	Medium
28	school	Anna Nehru Matriculation School	11.216056, 78.164447	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 11.2184729, 78.1635718



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 11.2164807, 78.1511632



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37868, 77.89498



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.23446, 78.15458



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.22243, 78.15379



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.22081, 78.15944



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.21910, 78.15911



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.21847, 78.16357



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.21841, 78.16359



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.21648, 78.15116



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.21655, 78.15114



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.21765, 78.15430
